

1. Administrative & Study Identification

Public Study Title: A Clinical Study of Intismeran Autogene (V940) Treatment and Pembrolizumab in People With Bladder Cancer (V940-005/INTERpath-005) **Full/Official Study Title:** A Phase 1/2 Study of V940 Plus Pembrolizumab With or Without Enfortumab Vedotin in Muscle-Invasive Urothelial Carcinoma (MIUC) (INTERpath-005) **Short Title/Acronym:** V940-005 / INTERpath-005 **Protocol Number:** V940-005 (IRB# IRB24-0053) **NCT Number:** NCT06305767 **Posting ID:** 381137111551453590 **Trial Status:** Recruiting **Sponsor/Company Name:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** Intismeran autogene (V940 / mRNA-4157), Pembrolizumab (Keytruda®), Enfortumab Vedotin (EV) **ATC Code (Pembrolizumab):** L01FF02 **EMA URL (Pembrolizumab):** [Keytruda EPAR](#) **Product Information Preferred UMLS Name:** Immunotherapy, Combination **Phase of Development:** Phase I/II **Medical Monitor:** Peter H. O'Donnell, MD (Lead Physician) / Merck Sharp & Dohme LLC

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of the perioperative regimen (Phase 1) and to evaluate Disease-Free Survival (DFS) per investigator assessment in the adjuvant setting (Phase 2). **Secondary Objectives:** To evaluate pathologic complete response (pCR) and pathologic downstaging (perioperative cohort); and to evaluate Overall Survival (OS), distant metastasis-free survival (DMFS), safety, and tolerability (adjuvant cohort).

2.2 Background & Rationale Researchers are seeking new ways to treat patients with high-risk muscle-invasive urothelial carcinoma (MIUC). Urothelial carcinoma is a type of bladder cancer that begins in cells that line the inside of the bladder and other parts of the urinary tract. V940 is an individualized neoantigen therapy (mRNA-4157) designed to treat each person's cancer by helping the patient's immune system identify and kill cancer cells based on specific proteins. This study aims to determine if administering V940 with pembrolizumab (with or without the antibody-drug conjugate enfortumab vedotin) can safely prevent MIUC from returning after radical surgery and improve survival outcomes compared to placebo and pembrolizumab.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled and Active-controlled (Adjuvant Cohort);

Single-arm (Perioperative Cohort) **Blinding:** Double-blind (Participant, Investigator) for Adjuvant Cohort; Open-label for Perioperative Cohort **Randomization:** 1:1 ratio (Adjuvant Cohort); N/A (Perioperative Cohort)

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 230 participants **Number of Sites:** ~61 sites globally (USA and 14 other countries) **Estimated Study Start:** 28-Mar-2024 **Phase 1 Cohort Start:** Enrollment of participants into the Phase 1 Perioperative Cohort is planned to start in approximately April 2025. **Estimated Primary Completion:** 23-Apr-2027 (Estimated Study Completion: 20-Oct-2031)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histological diagnosis of urothelial carcinoma (UC). 3. Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 2 assessed within 7 days before randomization. 4. Must provide blood samples per protocol to enable intismeran autogene (V940) production and circulating tumor DNA (ctDNA) testing. 5. Must provide a formalin-fixed paraffin-embedded (FFPE) tumor tissue sample for next generation sequencing (NGS). 6. **Adjuvant Cohort Specific:** Has MIUC; has high-risk pathologic disease after radical resection; for participants not receiving cisplatin-based neoadjuvant chemotherapy, ineligible to receive cisplatin according to protocol pre-defined criteria. 7. **Perioperative Cohort Specific:** Has muscle-invasive bladder cancer (MIBC); is deemed eligible for radical cystectomy (RC) and pelvic lymph node dissection (PLND) and agrees to undergo curative intent standard RC and PLND and neoadjuvant/adjuvant treatment per protocol; ineligible to receive cisplatin according to protocol pre-defined criteria.

4.2 Exclusion Criteria 1. Has received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. 2. Has known additional malignancy that is progressing or has required active treatment ≤ 3 years prior to study randomization. 3. Has current pneumonitis/interstitial lung disease. 4. Has active infection requiring systemic therapy. 5. Has active hepatitis B and hepatitis C virus infection. 6. **Adjuvant Cohort Specific:** Has received prior systemic anticancer therapy; has received prior neoadjuvant therapy, with the exception of neoadjuvant cisplatin-based chemotherapy for MIUC; has severe hypersensitivity to either intismeran autogene or pembrolizumab (MK-3475) and/or any of their excipients. 7. **Perioperative Cohort Specific:** Has received any prior systemic treatment, cancer vaccine treatment, chemoradiation, and/or radiation therapy treatment for MIBC; has severe hypersensitivity to either intismeran autogene, pembrolizumab, or EV and/or any of their excipients; has ongoing sensory or motor neuropathy; has active keratitis or corneal ulcerations.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Adjuvant Cohort:** V940 (1 mg Q3W for up to 9 doses) + Pembrolizumab (400 mg Q6W for up to 9 cycles). * **Perioperative Cohort:** Neoadjuvant Pembrolizumab (200 mg Q3W) + Enfortumab Vedotin (1.25 mg/kg Days 1 & 8 Q3W) + V940 (1 mg Q3W) before surgery; followed by Adjuvant Pembrolizumab + Enfortumab Vedotin + V940 (for a total of 9 doses of V940). **Route of Administration:** Intramuscular (IM) for V940; Intravenous (IV) for Pembrolizumab and Enfortumab Vedotin. **Duration of Treatment:** Up to 13 months for the Adjuvant Cohort; Up to 16 months for the Perioperative Cohort.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications to manage side effects (e.g., anti-emetics). **Prohibited:** Concurrent investigational therapies, other anti-neoplastic drugs, or live/live-attenuated vaccines within 30 days before first dose and during the treatment period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to -28 Days	Informed Consent, Blood/Tissue sampling for V940 manufacturing and NGS, ctDNA evaluation, Baseline Imaging
Baseline	Day 0	Randomization (Adjuvant Cohort) / Allocation (Perioperative Cohort), First Dose
Interim	Every 3 or 6 Weeks	Study drug administration (IM/IV), Safety Labs, Efficacy & Radiographic Assessments
End of Study	Month 13 to 16	Final Treatment Visit, Exit Interview, Long-term survival & recurrence follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Adjuvant Cohort:** Disease Free Survival (DFS) – Defined as the time from randomization until death from any cause, or presence of disease per investigator assessment with muscle-invasive ($\geq$ pT2) disease or any high-grade non-muscle invasive disease in the urothelial tract. 2. **Perioperative Cohort:** Number of Participants Who Experience an Adverse Event (AE). 3. **Perioperative Cohort:** Number of Participants Who Discontinue Study Treatment Due to an Adverse Event (AE).

7.2 Secondary & Exploratory Endpoints 1. **Adjuvant Cohort:** Overall Survival (OS). 2. **Adjuvant Cohort:** Distant Metastasis-Free Survival (DMFS). 3. **Adjuvant Cohort:** Number of Participants Who Experience an AE. 4. **Perioperative Cohort:** Pathologic Complete Response (pCR) and pathologic downstaging (Exploratory/Secondary).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via regular clinical evaluations, recording any untoward medical occurrence temporally associated with the use of study treatment. There is a specific focus on immune-related adverse events (irAEs) and EV-related toxicities (e.g., severe skin reactions, neuropathy).

Serious Adverse Events (SAEs): Required reporting to the sponsor within 24 hours of site awareness. **Data Monitoring Committee (DMC):** External, independent Data Monitoring Committee evaluating unblinded safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary efficacy outcomes (DFS, OS); Safety Set for AE/SAE evaluations. **Sample Size Justification:** N=230, powered appropriately to detect a statistically significant increase in DFS comparing the V940 + pembrolizumab arm versus the placebo + pembrolizumab arm. **Handling of Missing Data:** Standard censoring rules for time-to-event survival data (Kaplan-Meier estimates).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite and remote monitoring visits by the sponsor to ensure protocol compliance and data integrity.

Audit Trail: Robust Electronic Data Capture (EDC) system with full audit trail tracking and data clarification processes.

11. Study Identifiers & Governance

Primary ID: V940-005 **Registry Number:** NCT06305767 **Posting ID:** 381137111551453590 **Sponsor/Lead Organization:** Merck (Merck Sharp & Dohme LLC) **Study Title:** INterpath-005 **Official Regulatory Title:** A Phase 1/2 Study of V940 Plus Pembrolizumab With or Without Enfortumab Vedotin in Muscle-Invasive Urothelial Carcinoma (MIUC)

12. Clinical Framework (Bladder Cancer Specific)

Disease Area: Bladder Cancer **Preferred Name:** Malignant neoplasm of urinary bladder **Primary Condition:** Muscle-Invasive Urothelial Carcinoma (MIUC) / Muscle-Invasive Bladder Cancer (MIBC)

Diagnosis Threshold: Histologically confirmed high-risk MIUC after radical resection or prior to planned cystectomy (RC & PLND). **Medical Methodology:** Individualized neoantigen therapy

paired with PD-1 inhibitor and/or Antibody-Drug Conjugate (ADC).
Optimization Target: Disease-Free Survival, prevention of distant metastasis, and overall survival.

13. Study Procedures & Methodology

Management Pathway: Standard oncological pathway for radical cystectomy (RC) and pelvic lymph node dissection (PLND).

Education Protocol (HCP): Site training on V940 (IM) and Pembrolizumab/EV (IV) administration, sequence timing, and management of immune-related AEs. **Education Protocol (Patient):** Informed consent details regarding personalized vaccine manufacturing timelines and adherence to strict visit schedules.

Quality Audit Mechanism: Centralized independent review of radiographic imaging and pathology.

14. Observation & Follow-Up Schedule

Total Duration: Up to ~7 years (End of Study estimated 2031)

Onsite Visit Cadence: Every 3 weeks (Perioperative) or Every 6 weeks (Adjuvant) during active treatment phase. **Remote Monitoring Frequency:** Routine safety follow-ups as determined by protocol.

Checklist Review Interval: Pre-dose laboratory and physical assessments prior to each treatment cycle.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]				
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]				